

IoT Guardian Pro — Technical Documentation

Interactive Security Breach Simulator & Network Analytics Dashboard

This document provides a comprehensive technical overview of the **IoT Guardian Pro** simulation system located at **E:/PROJECTS/INTERN/PROJECT_ONE/breach-simu**.

1. Executive Summary

IoT Guardian Pro is an educational cyber-range and security simulator. It runs automated, harmless attack simulations (DDoS spikes, Dictionary Brute-Force, and Phishing) to teach administrators how anomalies look on network charts. It supports live telemetry via WebSockets and exports PDF/Excel incident reports.

2. Platform Architecture

The project has four distinct components:

1. **FastAPI Backend (backend/)**: Handles database entries, logs match history, manages SQLite states, and executes WebSocket network simulations.
2. **React Dashboard (frontend/)**: Displays real-time packet speeds, attack vectors, system stats, and generates audit reports.
3. **Landing Page (landing-page/)**: Introducing corporate portal services.
4. **Phishing Target (phishing-site/)**: A mock login screen (**login.html**) to demonstrate phishing vector captures.

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3. Core Simulation Modules

A. DDoS Simulation

Simulates packet spikes. When activated:

- The backend increases simulated packets/second rates by 500%.
- Sends high-speed streams via **/ws** to the frontend.

- Charts display latency spikes and packet rate peaks.

B. Brute-Force Simulation

Runs a mock login dictionary attack:

- Ingests passwords from a trimmed **rockyou.txt** dictionary file.
- Simulates hundreds of invalid login attempts against the system.
- Triggers alert notifications once threshold rates are breached.

C. Phishing Simulation

- Hosts a mock authentication page.
- Records username/password credentials entered during simulation runs (safely logged inside SQLite).
- Teaches users about credentials harvesting vectors.

4. Reports Generation (PDF & Excel)

The dashboard allows exporting audit sheets:

- **PDF Audits:** Uses **reportlab** to design dynamic PDFs detailing incident timelines, attack categories, IP origins, and severity.
- **Excel Reports:** Uses **openpyxl** to write data rows of all simulated logs, allowing easy parsing by analyst teams.

5. Setup & Execution Instructions

Setup Backend

1. Go to backend:

```
cd E:/PROJECTS/INTERN/PROJECT_ONE/breach-simu/backend
```

2. Install libraries:

```
pip install -r requirements.txt
```

3. Run FastAPI:

```
uvicorn app.main:app --reload
```

Setup Frontend

1. Go to frontend:

```
cd E:/PROJECTS/INTERN/PROJECT_ONE/breach-simu/frontend
```

2. Install and launch:

```
npm install npm run dev
```